INCH-POUND
MIL-C-82891(OS)
2 September 1993

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 14 MOD 1

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the Mark 14 Mod 1 Impulse Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified the issues shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-W-572	Wood Preservative: Water-Repellent

MILITARY

MIL-P-116	Preservation, Methods of
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-P-15011	Pallets, Material Handling, Wood Post Construction, 4-Way Entry

Beneficial comments and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1377
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MIL-P-23312	Pallets, Material Handling, (For Ordnance Items) Mark 3 Mod 0, Mark 12 Mods 0 & 1
MIL-P-46610	Primers, Percussion, Styphnate and Chlorate Types, For Small Arms Ammunition

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-453	Inspection, Radiographic
MIL-STD-1168	Ammunition Lot Numbering
MIL-STD-1322	Unit Loads of Ammunition and Explosives for Domestic and Overseas Shipments
DOD-STD-2101	Classification of Characteristics

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL 1699AS114	Cartridge, Impulse, Mark 14 Mod 1

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

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CODE OF FEDERAL REGULATIONS

49 CFR 171-199 Transportation, Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should request the latest edition and supplements thereto.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this specification shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Conformance to documents. The Mark 14 Mod 1 Impulse Cartridge covered by this specification shall be manufactured in accordance with DL 1699AS114 and all documents listed thereon.

3.3 Material. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Primary components. For the purposes of this specification, the primer (Drawing 7645339), infallible propellant (Drawing 838AS100) and HPC-95 propellant (Drawing 838AS121) are defined as primary components (see 4.5.1).

3.5 Cartridge inspection.

3.5.1 Visual inspection. Cartridges shall be free of the following visible defects: dents, deep scratches, cracks, burrs, split or cracked lip in the crimped area, damaged "O" ring or closure disk, incomplete or improper sealant application or any other imperfections which could prevent entry into the firing chamber or adversely affect performance or resistance to moisture (see 4.6.1).

3.5.2 Primers. The primers shall be furnished from lots manufactured within two years or recertified within 12 months of installation. The primers shall be manufactured in accordance with Drawing 7645339, loaded with mixture 5061W and shall be from lots which were accepted in accordance with MIL-P-46610, Class 1 (see 4.6.2).

3.5.2.1 Radiographs. Primers shall be free of the following defects when examined in accordance with 4.6.3: missing or abnormally thin charge; charge broken, cracked or porous; cracked or off-center anvil or other displacement or deformed inert parts; anvil abnormally close to the primer cup dome; and all other observable defects in assembly.

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3.5.3 Leakage. Each cartridge shall be leak tested and shall not exceed a leakage rate greater than 1 X 10^{-5} cm^3 of gas or air per second when tested in accordance with 4.6.4.

3.5.4 Propellant charges. All propellant charges shall be as specified on Drawing 1699AS114. Charges shall be determined by weight rather than by volume measurements. All charges shall be dried at 120°F for three hours minimum and desiccated prior to loading (see 4.6.5).

3.6 Radiographic examination. Cartridges shall be free of imperfections in components or assembly as verified by radiographic examination (see 4.6.6).

3.7 Ballistic requirements. Cartridges shall meet the requirements of 3.7.1 through 3.7.3 (see 4.6.7 and 4.6.8).

3.7.1 Chamber pressure. The Lower (L) and Upper (U) limits of maximum pressure shall be as follows:

Temperature	Lower P_{max}	Upper P_{max}
-65, +70 & +160°F	3,500 psi	6,000 psi

3.7.2 Misfire. There shall be no misfires (see 4.6.8.2).

3.7.3 Hangfire. There shall be no hangfires (see 4.6.8.3).

3.8 Packaging, packing and marking. Quality conformance inspection shall be conducted in accordance with the inspection criteria on the applicable drawings referenced in Section 5.

3.9 Workmanship. Cartridges shall be free of the following defects: inadequate or illegible marking, incomplete crimps, loose closure, excessive sealant, damaged cases, swollen cases that may prevent entry into firing chambers or ineffective seals (see 4.6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspections set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the

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Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed as specified in the applicable test method of 4.6.

4.4 First article inspection. First article inspection shall be performed after award of contract and prior to production by a facility designated in the contract or order (see 6.2). A first article sample acceptable for environmental and function testing shall be defined as a sample which has met the requirements of 3.2 through 3.6, 3.8 and 3.9 of this specification. First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials shall be the same as used for regular production and shall be from the intended sources of supply.

4.4.1 Sample size. The first article sample shall consist of 94 cartridges.

4.4.2 Inspection routine. Eighty-four cartridges shall be expended in the tests listed below and ten shall be retained for investigative purposes. The 84 cartridges shall be tested as shown in Table I. The tests shall be conducted in accordance with MIL-D-21625 (Basic). The cartridges shall be equally divided and conditioned for a minimum of six hours at -65, +70 and +160°F together with 12 each, equally divided untreated cartridges and fired within 3 minutes after removal from the conditioning ovens in the test fixture of 4.6.7. All units shall fire and meet the requirements of 3.7. The criteria of acceptance is as stated in 4.6.8.

4.4.3 Failure. Failure of the sample to pass the above tests shall be cause for rejection of the first article sample.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the tests and examinations of Table II.

4.5.1 Lot size. Unless otherwise specified (see 6.2), the production lot size shall not be less than 181 nor greater than 800 cartridges. Primary components from only one lot shall be used in one lot of cartridges, however one lot of primary components may be used in more than one lot of cartridges.

4.5.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-105. The ballistic sample shall be selected in accordance with Table III. An additional 5 cartridges shall be randomly selected with the test sample for investigative purposes.

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TABLE I. First article inspection routine.

Sequence No.	Test/Inspection	Number Tested/Inspected per Group				
1	Visual	6	12	12	18	36
2	Radiographic	6	12	12	18	36
3	40 foot drop	6				
4	15G shock		12			
5	Vibration			12		
6	T & H cycling				18	
7	Radiographic		12	12		
8	Ballistic tests:					
	-65°F		4	4	6	12
	+70°F		4	4	6	12
	+160°F		4	4	6	12

TABLE II. Quality conformance tests and examinations.

Item	Nature of test	Requirement	Test	Classification
Prim Comp	Single lot	3.4	4.5.1	(M101)
Cart	Visual	3.5.1	4.6.1	(M102)
Primer	Age, Cert	3.5.2	4.6.2	(M103)
Primer	Radiograph	3.5.2.1	4.6.3	(M104)
Propt	Type & weight	3.5.4	4.6.5	(M105)
Cart	Leakage	3.5.3	4.6.4	(M106)
Cart	Radiograph	3.6	4.6.6	(M107)
Cart	Ballistics	3.7	4.6.7	(M108)
Cart	Preservation	3.8	Sect. 5	(M109)

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TABLE III. Ballistic test sample sizes.

Lot size range	Sample size	Retains
181 - 300	21	5
301 - 500	24	5
501 - 800	30	5

4.5.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 Tests and examination.

4.6.1 Visual inspection. Examine all completely assembled cartridges and reject those having any of the defects noted in 3.5.1.

4.6.2 Primer verification. The contractor shall provide verification that the primers to be used were manufactured, loaded and accepted in accordance with the requirements of 3.5.2 and were furnished from lots manufactured within the prescribed time limits.

4.6.3 Primer radiographs. Examine all primers by radiograph in accordance with MIL-STD-453 and reject those having any of the defects of 3.5.2.1. Primers shall be kept on boards or trays until defects have been identified and removed.

4.6.4 Leakage. Each cartridge in the first article sample and in the production lot test sample shall be leak tested in a dry gas leak tester. Cartridges which exhibit a leak rate in excess of 1×10^{-5} cm^3/sec of air or gas at a pressure differential of 1.0 ± 0.1 atmosphere shall be considered defective. Cartridges of the preproduction sample which exhibit a leakage defect shall be rejected. If one or more cartridges of the production lot sample exhibit a leakage defect the lot shall be screened 100% at contractor expense or the lot shall be rejected.

4.6.5 Propellant. The charges are to be weighed to the nearest milligram for each cartridge. The drying of the propellant is to eliminate the presence of any surface moisture formed during storage.

4.6.6 Radiographic examination. All assembled cartridges in the first article and production lot shall be radiographically examined in accordance with MIL-STD-453. Any observable imperfections in components or assembly of the cartridge shall result in the rejection of that cartridge. The cartridges shall be positioned for the most revealing exposure.

4.6.6.1 Serialization. All cartridges shall be identified with serial numbers prior to examination.

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The serial numbers shall be pen or ink stamped on the cartridges with approximately 1/8-inch high characters and shall be located apart from the other cartridge markings and in consecutive order beginning with 001 in each production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order and each radiograph shall carry a permanent identification of the cartridge displayed thereon. The cartridges shall be positioned on their sides for the most revealing exposure. The radiograph identification shall include the complete lot number as stamped on the cartridges, Contract number, drawing number and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the X-ray.

4.6.6.2 Inspection. Defective cartridges found by the vendor shall be clearly identified on the X-ray and removed from the lot.

4.6.7 Ballistic testing. After successfully completing the tests of 4.6.1 through 4.6.6, the test cartridges shall be equally divided and conditioned for 6 hours minimum to 18 hours maximum at each of the following temperatures: -65, +70 and +160°F. The temperature tolerance shall be ± 5°F and the cartridges shall be fired within 3 minutes after removal from the conditioning chamber. The cartridges shall be fired in a closed chamber vessel, Drawing 847AS819, having an internal volume, with an expended cartridge in place, of 5.61 ±0.05 cubic inches and adapted for measurement of pressure. If any cartridge fails to meet the requirement of 3.7, the lot shall be rejected (see 6.3.1).

4.6.7.1 Retest. There shall be no retest unless, after investigation, it is determined that the failure was due solely to the test equipment.

4.6.8 Criteria of acceptability.

4.6.8.1 Chamber pressure. The chamber pressure shall meet the requirements of 3.7.1 at the temperatures specified. Lot acceptability shall be determined in accordance with the applicable provisions of MIL-STD-414, Table A-2, Level IV, at an AQL of 0.04.

4.6.8.2 Misfire. The failure of any cartridge to fire shall result in rejection of the lot represented unless such failure is plainly attributable to faulty test equipment or procedure.

4.6.8.3 Hangfire. One or more hangfires shall result in rejection of the lot represented unless such a failure is plainly attributable to faulty test equipment or procedure. A hangfire is defined as a firing in which the elapsed time between primer strike, by the firing mechanism, and start of pressure rise exceeds 0.050 seconds.

5. PACKAGING

5.1 Preservation and packing. For definitions of packaging levels see 6.3.2.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. One cartridge shall be packaged in an inner container in accordance with the drawings listed on DL 491828. A tight pack is mandatory.

5.1.2 Level C. When directed by the procuring activity, those cartridges which are intended for

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expenditure in preproduction sample testing and for production lot acceptance testing shall be packaged in accordance with Method III of MIL-P-116 and with Code of Federal Regulations 49 CFR 171-199. The containers shall be Performance Oriented Packaging (POP) certified. A tight pack is mandatory.

5.2 Packing. For definitions of packing levels see 6.3.3.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Eight inner containers as described in 5.1.1 shall be packed in accordance with the drawings listed on DL 491829.

5.2.2 Marking. Marking of the can and outer container shall be in accordance with DL 491828 and 491829, respectively, using appropriate nomenclature (Mark 14 Mod 1). The lot number shall conform to MIL-STD-1168. Special marking shall be in accordance with MIL-STD-129.

NOTE: The provisions of 5.2 are applicable to both air and surface shipments, with the exception of Parcel Post shipments, which may be packed in weather resistant fiberboard containers.

5.3 Palletization.

a. When the volume of all the shipping containers exceeds 30 cubic feet, they shall be palletized. The wooden pallets shall conform to MIL-P-15011 Style I (40" X 48") or Style IA (35" X 45½") and the steel pallets shall conform to MIL-P-23312, Mk 3 or Mk 12 Mods.. Each palletized unit load shall not exceed 2200 pounds. Pallets shall be uniform using fillers, battens and spacers to create a stable load evenly distributed. Boxes that are partially loaded shall be painted orange. Pallets shall be stabilized using steel strapping and corner braces. For the general requirements of palletization, MIL-STD-1322 applies.

(1) A maximum overhang of 2 inches is allowed for both length and width.
(2) The total height including pallet shall not exceed 52 inches.

b. Palletized loads of dangerous material shall be made up to provide easy accessibility to any one container in case of emergency.

c. All wood container blockings, spacers, battens and pallets shall be treated with wood preservative conforming to TT-W-572, Composition A.

d. The outside packaging, i.e., pallet crate, etc., shall contain only items from the same lot. The pallet once packed shall be totally stable before strapping the containers to the pallet.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The Mark 14 Mod 1 impulse cartridge is used to unlock and remove the canopy

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from an aircraft prior to ejection of the seat in an emergency escape. This cartridge is used in all series A-7 (except TA-7C) and all series F-8 aircraft.

6.2 Acquisition documents. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Issue of DoDISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1)

c. Whether first article inspection is required and, if so, specify the test activity (see 3.1, 4.4 and 6.4)

d. Production lot size and test activity (see 4.5.1)

e. Level of preservation and packing (see 5.1).

f. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material," DoD 4145.26M, are applicable.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP5, are applicable.

6.3 Definitions.

6.3.1 Hangfire. A hangfire is defined as a firing in which the elapsed time from primer strike, by the firing mechanism, and the start of pressure rise that exceeds the specified upper limit set for that performance time.

6.3.2 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and worldwide redistribution.

6.3.2.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.3.3 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and worldwide redistribution.

6.3.3.1 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

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6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractors current inventory, and the quantity of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bids unless specifically requested to do so in the solicitation.

6.5 Subject term (key word) listing.

Canopy ejection
Ballistic device

Preparing activity:
Navy - OS
(Project 1377-NE52)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-C-82891 (OS)	**2. DOCUMENT DATE (YYMMDD)** 930902

3. Cartridge, Impulse, Mark 14 Mod 1

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER, INDIAN HEAD DIVISION	b. TELEPHONE *Include Area Code)* (1) Commercial (301) 743-4700 X1958 (2) AUTOVON 354-4700 X1958
c. ADDRESS *(Include Zip Code)* NAVAL SURFACE WARFARE CENTER (CODE 8410T) 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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